CSE221 Assignment 08 Spring 2026

A. Friendship

1 second, 1024 megabytes

Study material: http://www.shafaetsplanet.com/?p=763

There is a group of $\mathbf{N}$ people living in a small village. Each person in the village has a unique identity — labeled with an integer value between 1 and N. Initially, the villagers don't have any friends. As time passes, they begin to form friendships.

In this problem, you will be given information of $\mathbf{K}$ friendships. You have to print an integer value that denotes the size of their friend circle.

Suppose five people are living in the village, labeled 1, 2, 3, 4, and 5. Initially, the size of each friend circle is one, since no friendships have been formed yet. One day, person 1 and person 2 become friends. So, the size of their friend circle becomes two. The next day, person 3 and person 4 become friends, and the size of their friend circle becomes two as well. After a few days, person 1 and person 4 become friends. Now, the size of their combined friend circle becomes four, consisting of persons 1, 2, 3, and 4.

Input

The first line contains two integers $N, K\ (1 \le N, K \le 3 \times 10^5)$ — the total number of people, total number of friendship created.

The next K lines contain two integers $a_i, a_j (1 \le a_i, a_j \le N, a_i \ne a_j)$ — two people become friends.

Output

For each friendship formation, output a single integer on a new line representing the size of the friend circle that the two people belong to after becoming friends.

input

```
8 7
2 4
4 5
3 6
4 7
3 1
2 7
6 2
```

output

```
2
3
2
4
3
4
7
```

input

```
6 5
4 6
2 6
1 6
1 4
1 2
```

output

```
2
3
4
4
4
```

input

```
2 1
1 2
```

output

```
2
```

input

```
5 7
4 5
3 5
3 4
2 5
2 3
1 5
1 4
```

output

```
2
3
3
4
4
5
5
```

In input sample 1,

Initially, 8 people in the village do not know each other.

{1} {2} {3} {4} {5} {6} {7} {8}

— After person 2 and person 4 become friends:

{1} {2,4} {3} {5} {6} {7} {8}

The output is 2, since the size of the friends circle {2,4} is 2.

— After person 4 and person 5 become friends:

{1} {2,4,5} {3} {6} {7} {8}

The output is 3, since the size of the friends circle {2,4,5} is 3.

— After person 3 and person 6 become friends:

{1} {2,4,5} {3,6} {7} {8}

The output is 2, since the size of the friends circle {3,6} is 2.

— After person 4 and person 7 become friends:

{1} {2,4,5,7} {3,6} {8}

The output is 4, since the size of the friends circle {2,4,5,7} is 4.

— After person 3 and person 1 become friends:

{2,4,5,7} {1,3,6} {8}

The output is 3, since the size of the friends circle {1,3,6} is 3.

Since person 2 and person 7 are already in the same friend circle, nothing changes:

{2,4,5,7} {1,3,6} {8}

The output is 4, since the size of the friends circle {2,4,5,7} is 4.

— After person 6 and person 2 become friends:

{2,4,5,7,1,3,6} {8}

The output is 7, since the size of the friends circle {2,4,5,7,1,3,6} is 7.

B. Help the King!

1 second, 1024 megabytes

In the kingdom of Beluga, there are $\mathbf{N}$ cities connected by $\mathbf{M}$ bidirectional roads. Each road has a maintenance cost. There is at least one way to travel between any two cities.

The king is worried about the growing cost of maintaining all these roads. To fix this, he asks his advisors for help.

The council suggests keeping only the roads needed to connect all the cities with the least total maintenance cost. Instead of building new roads, the king decides to save money by removing some of the existing ones.

Since you're known for your programming skills, the king calls on you. He wants you to figure out the minimum total maintenance cost that can be achieved by removing some roads—while still making sure that all the cities remain connected.

Input

The first line contains two integers $N, M (2 \le N \le 2 \times 10^5, 1 \le M \le 3 \times 10^5)$ — the number of vertices, total number of edges.

The next M lines will contain three integers $u_i, v_i, w_i (1 \le u_i, v_i \le N, 1 \le w_i \le 10^6)$ — there is an edge between the node u_i and the node v_i with a maintenance cost w_i.

Output

The output should contain a single integer, with the minimum total maintenance cost achievable.

input

```
4 3
3 4 5
3 1 5
2 1 2
```

output

```
12
```

input

```
6 5
2 6 3
2 3 3
2 1 4
6 5 1
5 4 2
```

output

```
13
```

input

```
2 1
1 2 9
```

output

```
9
```

input

```
5 7
1 2 8
1 4 3
1 5 4
2 4 8
2 3 5
4 3 4
3 5 5
```

output

```
16
```

C. Again MST

1 second, 256 megabytes

You are given a **bidirectional weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N. The graph contains no self-loops or multiple edges.

Your task is to find the total cost of the second-best Minimum Spanning Tree. If no such tree exists, print -1.

Note: The second-best MST must have a total weight strictly greater than that of the best (minimum) MST.

Input

The first line contains four integers $N, M (2 \le N \le 10^3, 1 \le M \le 2 \times 10^3)$ — the number of vertices, total number of edges.

The next M lines will contain three integers $u_i, v_i, w_i (1 \le u_i, v_i \le N, 1 \le w_i \le 10^6)$ — there is an edge between the node u_i and the node v_i with a weight w_i.

Output

Output the total cost of the second-best Minimum Spanning Tree. If no such tree exists, print -1.

input

```
6 7
1 2 1
2 3 2
3 1 3
1 4 5
4 5 4
5 6 5
6 4 5
```

output

```
18
```

input

```
5 5
1 2 3
2 3 4
3 4 5
4 5 1
5 1 2
```

output

```
11
```

D. Mission Unreachable

1.5 seconds, 256 megabytes

You are given an **undirected weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N. The graph contains no self-loops or multiple edges.

You are also given two nodes, a source S and a destination D. Now you have to remove some edges so that it is no longer possible to reach D from S.

The removal process works as follows: choose a value K and remove every edge whose weight is less than or equal to K. Your task is to find the minimum possible non-negative integer value of K that makes D unreachable from S.

Input

The first line contains a single integer T $(1 \le T \le 10^5)$ the number of test cases. Each of the test case contains four lines.

The first line contains two integers N, M, S and D $(1 \le N \le 10^5, 1 \le M \le 10^5, 1 \le S, D \le N, S \ne D)$ — the number of nodes, the total number of edges, the source and the destination.

The second line contains M integers $u_1, u_2, u_3 \ldots u_m$ $(1 \le u_i \le N)$ — where the i-th integer represents the node that is one endpoint of the i-th edge.

The third line contains M integers $v_1, v_2, v_3 \ldots v_m$ $(1 \le v_i \le N)$ — where the i-th integer represents the node that is other endpoint of the i-th edge.

The fourth line contains M integers $w_1, w_2, w_3 \ldots w_m$ $(1 \le w_i \le 10^8)$ — where the i-th integer represents the weight of the i-th edge.

For each test case, the i-th edge of the graph is an undirected edge between the i-th value in the second line and the i-th value in the third line.

It is guaranteed that the sum of N and M over all test cases is at most 2×10^5.

Output

For each test case, print a single non-negative integer — the minimum value of K such that after removing all edges with weight less than or equal to K, there is no path from S to D.

input

```
1
4 5 3 4
1 2 1 4 2
3 3 4 3 4
3 4 5 6 7
```

output

```
6
```

input

```
2
3 1 1 3
3
2
5
3 1 2 3
2
3
5
```

output

```
0
5
```

E. Task Selection I

1 second, 256 megabytes

You are a busy person with lots of tasks to do. You have a schedule of tasks represented by intervals of time, where each interval represents a task that you need to complete. However, you can only work on one task at a time, and you want to complete as many tasks as possible.

Given a list of N intervals of time, your task is to determine the maximum number of tasks you can complete and which tasks they are.

Input

The input consists of a single integer N $(1 \le N \le 10^5)$ representing the number of available tasks, followed by N lines for each of the task. The i th task is represented by two space-separated integers S_i and E_i ($0 \le S_i \le E_i \le 10^9$), the start and the end time of the task respectively.

Output

Output a single integer k, the maximum number of tasks you can complete, followed by k lines describing the k tasks you can complete.

If there are multiple solutions with the same maximum number of tasks, print any one of them.

input

```
6
1 3
2 5
3 7
4 6
6 8
7 9
```

output

```
3
1 3
4 6
7 9
```

input

```
5
1 4
2 5
6 7
4 8
3 6
```

output

```
2
1 4
6 7
```

input

```
7
0 4
3 4
1 5
9 10
6 9
2 3
1 2
```

output

```
3
1 2
3 4
6 9
```

F. Task Selection II

1 second, 256 megabytes

Given N tasks and M people, where each task has a start time and end time, implement a greedy algorithm to find the maximum number of tasks that can be completed by M people.

Each task can only be completed by one person and a person can only be assigned one task at a time. Two tasks cannot be completed simultaneously by the same person.

Input

The first line of the input contains T ($1 \le T \le 100$), the number of test cases. For each test case, first line of the input consists of two integers N and M ($1 \le N, M \le 10^3$), the number of activities and the number people respectively. This is followed by N lines representing the activities. Each line contains two integers S_i and E_i ($0 \le S_i \le E_i \le 10^9$) representing the start and end time of the activity respectively. Finally, note that the sum of N over all test cases and the sum of M over all test cases will both be at most 2000.

Output

For each test case, output a single integer representing the maximum number of activities that can be completed.

input

```
1
5 2
1 5
3 6
2 5
8 10
6 9
```

output

```
4
```

input

```
2
5 2
1 4
2 5
6 7
4 8
3 6
6 2
1 5
4 10
8 17
12 15
9 11
14 18
```

output

```
3
5
```

input

```
2
5 2
1 10
2 10
6 7
4 8
3 6
8 3
5 7
2 4
6 8
8 10
1 3
7 9
3 5
2 6
```

output

```
2
7
```

G. Tasks and Deadlines

1 second, 256 megabytes

You have to process n ($1 \le n \le 2 \cdot 10^5$) tasks. Each task has a duration and a deadline, and you will process the tasks in some order one after another. Your reward for a task is $d - f$ where d is its deadline and f is your finishing time. (The starting time is 0, and you have to process all tasks even if a task would yield negative reward.)

What is your maximum reward if you act optimally?

Input

The first input line has an integer n: the number of tasks.

After this, there are n lines that describe the tasks. Each line has two integers a and d ($1 \le a, d \le 10^6$): the duration and deadline of the task.

Output

Print one integer: the maximum reward.

input

```
3
6 10
8 15
5 12
```

output

```
2
```

input

```
2
2 2
1 10
```

output

```
8
```

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